ACTIVATION DISTANCES AND EXTREMAL FIBRES OF CONFLICT-TRIGGERED CYCLIC INCREMENTS

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ABSTRACT. We determine the forward dynamics and time-one inverse of a synchronous colour update on an arbitrary finite simple undirected graph: a vertex increments modulo $q \geq 3$ exactly when a neighbor has the same colour. Equal-colour edges persist, so each vertex is stationary until activation and advances forever afterwards. We prove that its first activation time is a shortest-path distance from the initial conflicts, with directed edge weights given by initial colour differences. This yields every iterate, the recurrent core and the sharp maximum entrance time $(q-1)(n-2)$ for $n \geq 3$. Independently, all predecessors of a target are reconstructed by binary masks satisfying a total-cover condition and directed predecessor closure. Using a self-contained static bound, we show that the largest time-one fibre over all n-vertex graphs and targets is $2^{n-1}-1$ for $n \geq 4$, attained exactly by constant targets on stars. The three-vertex exception is a constant target on a triangle, with four predecessors. The inverse characterization does not assume efficient general cover counting.

1. The update and its scope

Let $G = (V, E)$ be a finite simple undirected graph, $n = |V| \geq 0$, and $q \geq 3$. For $x \in (\mathbb{Z}/q\mathbb{Z})^V$, define the simultaneous update

$$F(x)_v = x_v + \mathbf{1}\{\text{some } u \sim v \text{ has } x_u = x_v\} \pmod q. \tag{1.1}$$

Isolated vertices hold. We call an equal-colour edge a conflict and its endpoints active. All tests in (1.1) use the old state. The entrance $h(x)$ is the least $t \geq 0$ for which $F^t(x)$ is periodic. Unlike a proper-colouring algorithm, this update never removes a conflict: both endpoints increment together. The issue is how the resulting clocks activate stationary vertices, and which old conflict sets can produce a given target.

The conflict-detection model of Motskin et al. [4, Section II-A] already allows a local function of the current colour and one conflict bit; (1.1) is one deterministic choice in that model. Their Algorithm 1 uses randomized recolouring. We do not claim a new information model or transfer that algorithm's convergence result. The successor-colour trigger in the cyclic cellular automaton differs from our equality trigger [2, Section 1]: on a monochromatic edge it holds, whereas (1.1) rotates. This literal distinction is not a claim excluding every possible enriched representation.

We give two explicit descriptions of (1.1). The first uses initial-colour directed arrival distances to determine every forward coordinate, exact entrance and recurrent action. The second uses target colours to characterize all active masks in a one-step inverse and to find its uniform extremizers. Shortest paths and binary reconstruction are standard tools. The inverse uses the established class of 2-total vertex covers [1]; general total-cover counting already has complexity results [3, Corollary 2.5]. Our elementary static bound below is supporting material, not a separately claimed new graph-counting theory. Section 2 proves the forward formula and Section 3 proves the inverse statements.

2. Exact activation distances

Let $S(x)$ be the endpoints of the initial conflicts. Give both directed versions of each edge the weights

$$w_x(u, v) = (x_v - x_u) \bmod q \in \{0, \ldots, q-1\}.$$

Write $d_x(v)$ for the minimum total weight of a directed path from $S(x)$ to v, and set it to ∞ if no such path exists. Paths of length zero are allowed. A maximum over no finite distances means zero.

Theorem 2.1. *The first conflict time of v is $d_x(v)$. For every integer $t \geq 0$,*

$$F^t(x)_v = \begin{cases} x_v + \max\{0, t - d_x(v)\} \pmod q, & d_x(v) < \infty, \\ x_v, & d_x(v) = \infty. \end{cases} \tag{2.1}$$

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Moreover, $h(x) = \max_{d_x(v)<\infty} d_x(v)$*. A state is recurrent exactly when each connected component is either properly coloured, hence fixed, or has a same-coloured neighbor at every vertex, hence increments globally. Every nonfixed recurrent orbit has exact period* q*. The maximum entrance over all graphs and sources on* n *vertices is*

$$H_q(n) = \begin{cases} 0, & n \le 2, \\ (q-1)(n-2), & n \ge 3. \end{cases}$$

Proof. An active edge remains active after an update, since its two colours increment equally. Thus the active set only grows. If $\tau(v)$ is the first conflict time, then v is stationary through time $\tau(v)$ and increments at each subsequent update. This gives (2.1) with τ in place of d_x.

Suppose u first activates at time $a < \infty$. Its colour at that time is x_u. Unless a neighbor v activates earlier, its colour is still x_v, and it meets the advancing colour of u at time $a + w_x(u, v)$. Consequently $\tau(v) \le \tau(u) + w_x(u, v)$. A zero-weight edge is initially monochromatic, so both its endpoints already have time zero; it cannot spontaneously activate an unseeded region. The edge inequality along every seed path proves $\tau(v) \le d_x(v)$ whenever the latter is finite.

Conversely, suppose $\tau(v) = s > 0$. A neighbor u meeting v at time s must have activated earlier. Indeed, if both first activated at s, both would still have their initial colours, making their equality an initial conflict. With $a = \tau(u) < s$, the first meeting of that advancing color with the stationary v occurs after exactly $w_x(u, v)$ steps: an earlier congruent meeting would already activate v. Hence $s = a+w_x(u, v)$. Repeating backwards strictly decreases activation times until reaching a seed and yields a path of total weight s. Thus $d_x(v) \le \tau(v)$. This also excludes finite activation in a seedless component, and proves the first assertion including infinite distances.

Every vertex in a seeded component has finite distance. At the last such arrival all its vertices are active and hence advance together forever; unseeded components are proper and hold forever. Before the last arrival, the active set will strictly increase at a future time, so monotonicity excludes periodicity. These observations prove the exact entrance and the stated recurrent core. If any component advances, a return after p steps forces $p \equiv 0 \pmod q$ at an advancing vertex, proving its exact period.

A seeded component contains at least two seeds. A minimum-weight path to a nonseed can be chosen simple and with no seed after its starting vertex. It therefore has at most $n-2$ edges, each of weight at most $q-1$. For $n \le 2$ there is no seeded component containing a nonseed, giving height zero. For $n \ge 3$ take the path $0 - 1 - \cdots - (n-1)$ with $x_0 = x_1 = 0$ and $x_i = -(i-1) \pmod q$ for $i \ge 2$. Only the first edge is initially monochromatic. The unique simple seed route to vertex $n-1$ has $n-2$ forward edges of weight $q-1$; deleting any backtracking cannot increase a path's weight. Its distance is $(q-1)(n-2)$, attaining the bound. ☐

3. A target decoder and sharp fibres

For a target y, let H_y be the spanning graph of its monochromatic edges. Let D_y have an arc $u \to v$ when $u \sim v$ and $y_v = y_u + 1 \pmod q$. A vertex cover whose induced graph has no isolated vertex is called a 2-total cover; we allow the empty cover. In particular, isolates of H_y must be outside such a cover.

Theorem 3.1. *The predecessors of* y *are, uniquely,* $x_v = y_v - \mathbf{1}_A(v) \pmod q$*, where* $A \subseteq V$ *satisfies:*

(1) A *is a vertex cover of* H_y*;*
(2) *every vertex of* A *has a neighbor in* A *in* H_y*;*
(3) *an arc* $u \to v$ *in* D_y *with* $v \in A$ *implies* $u \in A$*.*

Proof. Let A be the advancing set of any predecessor. Its old colours are necessarily $y - \mathbf{1}_A$. Every old equal-colour edge has both endpoints in A. For two endpoints in A, old equality is exactly target equality; each vertex in A therefore needs an internal H_y neighbor. Two endpoints outside A cannot be equal in the target, since both held. Finally, when $u \notin A$ and $v \in A$, old equality is equivalent to $y_v = y_u + 1 \pmod q$, which would contradict u holding. These are the three conditions. Conversely they ensure that the endpoints of the old equal-colour edges in $y - \mathbf{1}_A$ are exactly A. Thus precisely these vertices advance and the image is y. Different masks give different old states, completing the bijection. ☐

Write $T(H)$ for the number of 2-total covers of a simple graph H, with our empty-cover convention. Equivalently it counts independent sets I for which $H - I$ has no isolated vertex. The decoder gives

$$|F^{-1}(y)| \le T(H_y). \tag{3.1}$$

For a constant target, $H_y = G$ and D_y has no arcs, so equality holds. We include the following elementary static support proof to make the dynamical extremum self-contained.

Lemma 3.2. *For a graph H on $n \geq 4$ vertices, $T(H) \leq 2^{n-1} - 1$, with equality exactly for a star $K_{1,n-1}$. On three vertices, $T(H) \leq 4$, with equality exactly for a triangle.*

Proof. For a star on $k \geq 3$ vertices the covers consist of the centre and any nonempty set of leaves, giving $2^{k-1} - 1$. A single edge has one such cover; on three connected vertices the path has three and the triangle four.

A connected nonstar with at least four vertices contains a four-vertex path as a subgraph. To see this, if a longest path were a, b, c, any further vertex can attach only to b: attachment to a or c, or a longer route from the path, extends it. An edge between leaves, including a, c, would also create a four-vertex path. The graph would therefore be a star.

Now suppose a connected nonstar has $k \geq 5$ vertices. Choose a four-vertex path and an edge uv from a vertex u of the path to a vertex v outside it. The path has eight independent sets, at least two containing each chosen u. Ignoring other edges gives $8 \cdot 2^{k-4}$ independent sets; the extra edge forbids at least $2 \cdot 2^{k-5}$ of them. Hence

$$T(H) \leq i(H) \leq 7 \cdot 2^{k-4} < 2^{k-1} - 1,$$

where $i(H)$ is the number of independent sets and the strict inequality uses $2^{k-4} \geq 2$. For $k = 4$, the connected graphs are the star, path, paw, four-cycle, diamond and complete graph. Listing independent complements with no isolated vertex left gives, respectively, $7, 4, 6, 5, 6, 5$. Thus only the star attains seven.

The count T multiplies over components, and an isolate contributes one. A nontrivial component of size k has $T \leq 2^{k-1}$. Two or more nontrivial components therefore give at most $2^{n-2} < 2^{n-1} - 1$. One such component with isolates has strictly smaller order and again gives a strict bound, including the triangle count four. An edgeless graph has $T = 1$. The three-vertex disconnected cases also have $T = 1$. These cases complete the proof and the equality classification. □

Theorem 3.3. *For any $q \geq 3$, the maximum time-one fibre over all graphs and targets on n vertices is*

$$M(n) = \begin{cases} 1, & n \leq 2, \\ 4, & n = 3, \\ 2^{n-1} - 1, & n \geq 4. \end{cases}$$

For $n = 3$ the extremizers are exactly constant targets on triangles, and for $n \geq 4$ exactly constant targets on stars.

Proof. Combine (3.1) with Lemma 3.2. At equality H_y is a spanning star, or a triangle when $n = 3$. Since this monochromatic graph is connected, y is constant on all vertices. Then $H_y = G$, so no extra nonmonochromatic edges are possible. Conversely every constant target on the claimed graph attains $T(G)$. For $n \leq 2$, an edgeless graph has the identity update; on a single edge, unequal colours hold and equal colours rotate together. This is a permutation and every fibre has size one, including the empty state. □

4. Scope and reproducibility

The forward formula holds at every time, but the inverse concerns one step. No efficient arbitrary-graph cover counter, all-time inverse or complete basin/recurrent-state census is asserted.

The accompanying standalone `verify.py` compares literal finite orbits with weighted distances and every target's exact source set with the mask decoder. It covers all graphs through four vertices at $q = 3, 4, 5$, all static graphs through six vertices, and the sharp paths for $2 \leq n \leq 20$, $3 \leq q \leq 9$. Its complete `CANONICAL.json` records 1,029,769 assertions. These finite checks pressure the deductions; the all-parameter results follow from the proofs above.

References

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